

Learned-Geometry Confirmation of Concern-Gated Retrieval: the COGR-E2b 3x3 Crossed Design

Program: Concern-Gated Retrieval (COGR) – Wave 1b (COGR-E2b) **Package:** experiments/concern_gated_retrieval_e2/wave1b/ **Predecessors (imported, frozen, never edited):** experiments/concern_gated_retrieval_e2/wave0/ (Wave 0 hash 9683c5a1f4010361d6e120bcabd2743fb33e8cc9c7c79d5bd9b1d9f9f8889c23), experiments/concern_gated_retrieval_e2/wave1a/ (Wave 1a hash c23b31d977d7c169d57ca12cdfdbc8ad3a59188542efbdf802e341b1c8937209, screen decision KILL). **Date:** 2026-07-24 **Human director:** Jawaun Brown **Status:** technical report accompanying the Wave 1b preregistration ([PREREGISTRATION.md](../.. /experiments/concern_gated_retrieval_e2/wave1b/PREREGISTRATION.md)), the two non-compensatory promotion contracts ([PROMOTION_CONTRACT_L1.md](../.. /experiments/concern_gated_retrieval_e2/wave1b/PROMOTION_CONTRACT_L1.md), [PROMOTION_CONTRACT_L2.md](../.. /experiments/concern_gated_retrieval_e2/wave1b/PROMOTION_CONTRACT_L2.md)), and the crossed-factorial confirmatory receipt. The receipt is the load-bearing step; every numerical row in §5 that depends on the Modal L4 confirmatory run reports the completed confirmatory run (L1 KILL, L2 WITHHELD) and [PROVENANCE.md](../.. /experiments/concern_gated_retrieval_e2/wave1b/PROVENANCE.md) is populated. The paper is written **once** with two verdicts that are issued **separately**; the writing of the receipt – not the writing of the report – turns the placeholders into authoritative values or into an honest KILL.

Wave-boundary reminder. Wave 1b **can KILL** the L1 candidate mechanism (representation contribution) and it **can KILL** the L2 candidate rule (concern recovery + specificity) independently. Wave 1b **cannot** establish substrate transfer (L3), external-agent applicability (L4), a cognitive/self-model interpretation (L5), or semantic meaning. Those remain Wave 2+ objects. Any restatement of this paper as an L3, L4, L5, or selfhood claim is inconsistent with the two promotion contracts and is not authorized by this report.

Abstract

Wave 1b of the Concern-Gated Retrieval E2 program is the confirmatory **crossed learned-geometry × concern** experiment for the two claims the Wave 0 calibration + Wave 1a screen jointly made rejectable: **L1 representation contribution** (does the candidate mechanism – rarity-corrected joint context × concern retrieval – beat every matched-budget baseline on sealed task outcome when memory geometry is being *learned* or *frequency-matched random*, and survive an interventional edge-ablation causal test?) and **L2 concern recovery + specificity** (does the online concern-update rule recover from an adversarially wrong prior, beat every information-matched generic signal, and survive a split-budget uncertainty-audit ablation, *without being reproduced* by naïve exploration alone?). Wave 1b crosses Geometry ∈ {LEARNED, FREQ_MATCHED_RANDOM, ORACLE_WITHHELD} with Concern ∈ {FROZEN_WRONG, ONLINE_LEARNED_IPS, ONLINE_LEARNED_DR, ORACLE} with Family ∈ {delayed_commitments_v2, maintenance_fault_v2, resource_constrained_v2} for 27 non-oracle cells × 300 confirmatory seeds each (paired within family-slice on the Wave 0 confirmatory seed range 200000..201999).

The L1 verdict is scored on the FROZEN_WRONG concern rows crossed with non-ceiling geometry (LEARNED, FREQ_MATCHED_RANDOM). This isolates the representation-contribution question from the concern-recovery question. The L2 verdict is scored on the ONLINE_LEARNED concern rows crossed with LEARNED geometry, gated by a *pre-run family-design assertion* (§4.4) that refuses to score L2 if any generic-signal baseline reproduces the oracle top-k – the exact failure mode that KILLED Wave 1a. Both verdicts are non-compensatory over ten fatal gates (G0-G9), including SET-level oracle-regret metrics that catch missed combinatorial-utility structure (contradictory pairs, complementary pairs, dangerous conjunctions), a **statistical** leakage audit (label-permutation + randomized-generator controls) on the permitted graph features, an interventional edge-ablation test, and an adversarial single-source concern-poisoning stress inside the Wave 0 influence bound. Utility is task-based (Δ_{task}); epiplexity S^{φ} is a

dependent variable and optional bonus ($\beta = 0$ in the L1 and L2 gates), never the verifier.

The confirmatory Modal L4 run completed (27 cells $\times$ 300 seeds = 8 100 receipts; \$10.80 on 64 L4 workers). **The verdict is L1 KILL, L2 WITHHELD.** On all three redesigned families the candidate mechanism's learned-vs-random-geometry contrast is essentially zero (mean_delta = -0.022 / -0.005 / -0.003), so a frequency-matched random graph reproduces its performance and the learned geometry contributes nothing; and the interventional edge-ablation moves the sealed outcome in the predicted direction only 44–63% of the time (below the 0.70 causal requirement). Crucially, the statistical leakage audits **pass** on all three families (label-permutation $p = 0.37$ – 0.59 , far above the 0.01 tolerance), so – unlike Wave 1a, where recency was a covert oracle – these families carry no hidden shortcut. Non-ceiling headroom is healthy (~ 0.52 – 0.57), so the KILL is not a ceiling artifact. This is a **clean falsification** of the L1 dual-source-retrieval claim on honest learned geometry: once the fixture no longer leaks the answer, rarity-corrected joint context $\times$ concern retrieval does not beat matched-budget baselines. L2 is withheld (not KILLED) per the noncompensatory contract, so the concern-recovery question can be re-opened by a future wave that first establishes a passing L1. The KILL is reported honestly – no post-hoc threshold swaps, no corpus swaps, no unregistered replay knobs.

1. Background

1.1 The two-flashlight decomposition, restated one more time

A bounded agent knows more than it can hold in its active representation. Its problem is therefore not only how to store knowledge but how to decide which currently absent fact deserves scarce attention *now*. Concern-gated retrieval decomposes off-context recall into two beams that must intersect. One beam is **context** – what the currently active representation demands. The other beam is **concern** – what the agent's history says matters to it. A candidate becomes a retrieval nomination only where the two beams overlap [1, § "The intuition: two flashlights over memory"]. On the canonical birthday-style example the two beams are "today is October 4" and "my partner's birthday matters to me on the day-of-year." Neither beam is loud enough to fire alone: the context is absorbed with unrelated work, the concern is a low-rate but load-bearing preference. The overlap is a specifically off-context need. This is an **AND**, not an OR: retrieve what is relevant now *and* important to the agent, then test whether attending to it actually helps.

The decomposition raises three questions the L0 pilot could not adjudicate and Wave 0 and Wave 1a were staged to make rejectable:

concern-only baselines at matched information budget on a memory geometry that does not encode the answer? (L1.)

own history, or wholesale from experience under a wrong start – without leaking through evaluator-only fields? (L2.)

result cannot be relabelled as evidence for L2 and a KILL on L2 cannot invalidate L1? (Roadmap [1, § "Wave 1 – staged mechanism identification"] pins this as a non-compensatory boundary; Wave 1b implements it as two contracts, one per claim.)

- Does the *joint* nomination outperform the *context-only* and
- Can the concern beam be *learned* – either partially from the agent's
- Are the answers to (L1) and (L2) *separable*, so that a valid L1

Wave 1b is the crossed learned-geometry $\times$ concern experiment that adjudicates all three questions at once, with the safeguards Wave 1a's KILL taught us to install.

1.2 What the L0 pilot established (and what it did not)

The frozen L0 pilot at experiments/concern_gated_retrieval/ established that the two-flashlight decomposition can be made precise, that its numerical plumbing (weighted graph,

personalized PageRank, epiplexity filter, coincidence intersection) is implementable, and that on the authored graph family the composition discriminates registered synthetic roles under frozen seeds and regimes [3, Wave 0 report §1.3]. That is an L0 executable-diagnostic result and nothing more. The pilot could not adjudicate whether joint retrieval helps when geometry is learned or withheld (its graph *encoded* the answer), whether concern can be recovered from misspecification (all initial, learned, and oracle concern conditions saturated at hit@1 = 1.000), whether multiplicative intersection is necessary (the additive fusion tied the product in two of three regimes), whether semantic meaning or selfhood is present (not tested), or whether the mechanism has any real-agent bottleneck to solve (not tested).

1.3 What Wave 0 froze

Wave 0 was a calibration-and-scaffolding step [3]. Its promotable deliverables (all issued as *variance* estimates, not retrieval winners) are the per-family threshold row Wave 1 confirmatory rows are scored against:

delayed_commitments	mu_best = 0.5314	sigma_best = 0.0218	headroom = 0.4845	delta_thresh_L1 = 0.0484
maintenance_fault	mu_best = 0.5029	sigma_best = 0.0267	headroom = 0.4548	delta_thresh_L1 = 0.0534
resource_constrained	mu_best = 0.5750	sigma_best = 0.0250	headroom = 0.4291	delta_thresh_L1 = 0.0500

Wave 0 also froze: the sealed environment interface (SealedEnvironment exposing only observe and evaluate, evaluate firing at most once per episode, an IntegrityAudit AST walker that flags any policy that dereferences role, utility, or `_answer_key`); the template-split runtime tripwire (TemplateBucket $\in$ {CALIBRATION, CONFIRMATION} with default-deny and a LeakageError on any mixed access); the wrong-prior specification (alarm at `w_alarm_init` = 1.0, at least one true commitment suppressed to `w_commit_init` = 0.05, at least one uniform); the LoggedProbePolicy(`epsilon`=0.05) propensity-logging scaffolding under a 0.01 coverage floor; and the confirmatory seed range 200000..201999 disjoint from the calibration range 100000..100999. The Wave 0 analysis hash 9683c5a1f4010361d6e120bcabd2743fb33e8cc9c7c79d5bd9b1d9f9f8889c23 binds every one of these to the exact bytes of the calibration code.

Wave 0 explicitly did **not** update the wrong prior at evaluation time, did **not** learn the memory geometry, and did **not** perform the premise audit against governed real-world traces [3, §6]. Those three noncompensatory boundaries define the Wave 1 experiment surface.

1.4 What Wave 1a KILLED, and what Wave 1b inherits

Wave 1a (COGR-E2a) was a **screen** for the concern-update rule on *fixed* withheld geometry [2]. It crossed five conditions (frozen_wrong baseline, online_learned_ips, online_learned_dr, oracle_ceiling diagnostic, shuffled control, wrong_agent control) with the three Wave 0 families on the confirmatory seed range, wrapped every receipt-producing condition in LoggedProbePolicy(`epsilon`=0.05) so IPS/DR debiasing was well-defined, and ran a seven-gate non-compensatory promotion contract (G0-G7). The aggregate screen decision was KILL. Two failure modes emerged, both important to carry into Wave 1b honestly:

propensity-weighted coverage of the true commitment region was 0.000 on both online_learned_ips and online_learned_dr – the `epsilon` = 0.05 exploration budget could not reach the suppressed commitment region at all. Shuffled and wrong-agent controls also under-covered (0.066 and 0.059 respectively), well above the 0.01 floor but showing the same qualitative pattern.

1. G1 coverage collapse on delayed_commitments:

info_matched_recency reproduced the oracle ceiling *byte-for-byte* (0.5315 / 0.4772 / 0.6000). Because Wave 0's family generators placed the load-bearing memory at a family-consistent temporal offset, a recency-only ranker was, by construction, information-*more-than*-matched to the concern rule – it was receiving free access to the answer via a permitted feature. The G3 gate correctly refused to promote any composition that could not distinguish itself from that baseline. This is a *family-design* confound in Wave 0's fixture; it is not a fatal flaw in the two-flashlight decomposition and it does not block a Wave 1b L1 row that runs on redesigned families and on non-ceiling geometry.

2. G3 specificity failure on all three families:

Per the roadmap [1, § "Wave 1 – staged mechanism identification"] and the Wave 1a promotion contract, a Wave 1a KILL **withholds** L2 but does not block Wave 1b's L1 rows. Wave 1b runs L1 unconditionally; L2 rows run *iff* Wave 1b's redesigned families pass the pre-run recency-decoupling assertion (§4.4) and *iff* Wave 1b's L1 verdict fires PASS.

1.5 Two shortcuts E2b breaks

Wave 1b breaks the two shortcuts Wave 0 and Wave 1a preserved:

Shortcut 1: fixed geometry. Wave 0 and Wave 1a ran on the fixed withheld graph from `wave0.graph_learn.build_withheld_graph`. That graph's role labels are *not encoded* in its adjacency structure – it was withheld precisely so that the graph itself could not do the pattern matching – but the graph is *unchanged across episodes*. A mechanism that works on withheld geometry could still be trading on a specific graph shape. Wave 1b crosses LEARNED (per-episode audit-clean graph learning from policy-visible history), `FREQ_MATCHED_RANDOM` (a degree-preserving frequency-matched null over the same reference graph), and `ORACLE_WITHHELD` (the Wave 0 withheld graph, refused for promotion by `promotion_admit_geometry`). L1 is scored on the `FROZEN_WRONG` concern rows *only* on non-ceiling geometry (`LEARNED`, `FREQ_MATCHED_RANDOM`).

Shortcut 2: additive utility. Wave 0 and Wave 1a scored utility per-candidate additively: $\Delta(v) = J_g(v) - J_g(\emptyset)$. That aggregation made "useful bundles," "contradictory pairs," "complementary pairs," and "dangerous conjunctions" invisible – an allergy note that helps alone but combines with a nut memory and a birthday-cake memory to plan a *harmful* action would be scored as three positive singletons. Wave 1b promotes utility to a **first-class SET-level function** $\Delta_{\text{task}}(S) = J_g(\text{trajectory} \mid S) - J_g(\text{trajectory} \mid \emptyset)$ and evaluates the exhaustive oracle over all singletons, all pairs, and feasibility-gated triples per episode; §4.3 specifies the enumeration and the four decisive metrics (`oracle_recall_at_k`, `simple_regret_set`, `cumulative_regret`, `interaction_recovery`). The L1 gate scores Δ_{task} ($\beta = 0$ in the utility function); epiplexity $S^\emptyset$ is reported as a dependent variable diagnostic only.

2. Wave 0 summary (one page)

Wave 0 is a *calibration-only, scaffolding-only* step whose entire promotable claim is a signed preregistration, a non-compensatory seven-gate promotion contract (G0-G6), and a Modal L4 calibration receipt binding every threshold Wave 1 must clear. Its target object is the calibration variance estimate, not a retrieval winner [3, §2].

Substrate. A finite undirected weighted graph over typed nodes; imported unchanged from the L0 pilot. Personalized PageRank primitives are reused byte-for-byte so a Wave 1 tie between the candidate mechanism and the best matched-budget baseline cannot be blamed on a divergent PPR implementation.

Three procedural families. `delayed_commitments`, `maintenance_fault`, `resource_constrained`. Each instantiates the same abstract retrieval problem – "identify the off-context fact whose loading would improve the sealed outcome" – through different surface structure. Sealed reward is scalar in $[-1, +1]$; the load-bearing target's expected reward differential over the best distractor is at most 0.6 so no reasonable two-sided method starts at ceiling [3, §2.2, §6].

Sealed environment. Only `observe(episode) -> EpisodeContext` and `evaluate(choice) -> SealedOutcome`. `evaluate` may fire at most once per episode; a second call raises `SealedEvaluationError`. `EpisodeContext` is a frozen dataclass; attribute access outside its declared fields raises. Every evaluator-only field enumerated in Wave 0 preregistration §4.1 (`role`, `utility`, `_answer_key`, `oracle_concern`, `wrong_agent_id`, `template_family_split`, `paraphrase_family`, `generator_seed_kind`, `epiplexity_future_target`, `sealed_outcome_receipt`) is unreachable from any policy-visible code path.

Anti-leakage. Four layers: (1) the enumerated evaluator-only field list; (2) the sealed environment interface; (3) the static `IntegrityAudit` AST walker over every rank callable in

wave0.baselines; (4) the template-split runtime tripwire with default-deny and the environment guard COGR_WAVE0_CONFIRMATORY_RUN=1 required to touch confirmatory rows.

Wrong-prior specification. Alarm region at `w_alarm_init = 1.0`; at least one true commitment region suppressed to `w_commit_init = 0.05`; at least one other true commitment left at uniform. Per-family identifiers held only in the evaluator's private state.

Frozen thresholds. Populated from the Modal L4 calibration receipt into Wave 0 PREREGISTRATION.md §8; hash-bound by `WAVE0_ANALYSIS_HASH = 9683c5a1...`. The three per-family rows above (§1.3) are the exact values Wave 1b L1 confirms against.

Wave 0 does not test learned memory geometry, does not update the wrong prior, does not perform the premise audit, and does not license any L1, L2, or higher claim. Wave 1b inherits every one of Wave 0's frozen objects read-only and re-exports none of them.

3. Wave 1a summary (one page)

Wave 1a (COGR-E2a) is a *concern-recovery screen only* on fixed withheld geometry [2]. It targets exactly the composition `LoggedProbePolicy(epsilon=0.05) + update_concern(estimator ∈ {ips, dr}) + poisoning guard` – three off-the-shelf Wave 0 primitives that together specify the update rule Wave 1b's L2 gate needs.

Design. Five conditions × three families × up to 300 paired seeds per cell = 6320 receipts.

#	Condition	Concern state	Role
C1	frozen_wrong	Wave 0 wrong prior held fixed	Baseline
C2a	online_learned_ips	Online-updated via IPS	Candidate
C2b	online_learned_dr	Online-updated via DR	Candidate (2nd view)
C3	oracle	Oracle concern held fixed	Diagnostic ceiling (never promotable)
C4	shuffled	Wrong prior with anchor labels permuted	Specificity control
C5	wrong_agent	Concern profile from another agent	Specificity control

Every receipt-producing condition wrapped its nomination policy in `LoggedProbePolicy(epsilon=0.05)` so selection propensities were logged and were the sole quantity IPS/DR divided by. A pre-analysis coverage audit rejected any confirmatory row whose propensity-weighted coverage of the true commitment region fell below 0.01; per-family thresholds were 0.04845 / 0.05340 / 0.05000 on the paired-seed lower bound.

Seven fatal gates. G0 anti-leakage; G1 coverage; G2 propensity accounting; G3 specificity (must beat every info-matched generic signal AND the shuffled and wrong-agent controls by preregistered margins); G4 per-family effect; G5 seed independence; G6 code freeze; G7 Modal budget.

Verdict: KILL. The full aggregate reasons are enumerated in the Wave 1a screen receipt at `experiments/concern_gated_retrieval_e2/wave1a/results/verdict.json` under `aggregate_kill_reasons`. The two failure modes named in §1.4 above are the load-bearing ones:

both online arms – the suppressed commitment region was starved by the `epsilon = 0.05` exploration budget).

(`info_matched_recency` reproduced the oracle ceiling; on `delayed_commitments` the margin was -0.5562, i.e. the online concern rule was *below* the recency baseline by more than half the reward range, because recency was catching the load-bearing memory every seed by construction of Wave 0's family fixtures).

- G1 coverage collapse on `delayed_commitments` (coverage = 0.000 on

- G3 specificity failure on all three families

resource_constrained produced a genuine +0.226 effect for both online variants against every non-recency comparator – the online concern rule *does* recover useful priorities on that family – but the specificity gate correctly refused to promote a composition that could not distinguish itself from a recency baseline that had covert access to the answer.

What Wave 1a licensed for Wave 1b. Per the roadmap's noncompensatory rule and the Wave 1a promotion contract's demotion clause: a Wave 1a KILL withholds L2 but does not block Wave 1b's L1 rows. Wave 1b's L2 rows are additionally blocked-by the family redesign passing a pre-run recency-decoupling assertion (§4.4) – that is the mechanism by which the Wave 1a specificity KILL cannot silently recur under a family redesign that failed to actually fix the recency-oracle collision.

4. Design

4.1 The 3 × 3 × 3 crossed factorial

Wave 1b crosses three axes:

Geometry axis ∈ {LEARNED, *FREQ_MATCHED_RANDOM*, ORACLE_WITHHELD}. LEARNED – per-episode audit-clean graph learning over the policy-visible history and candidate set (`wave1b.learned_geometry.learn_graph`), reusing Wave 0's `apply_concern_warp` for the concern-weighted PageRank step. Never consults role labels, answer keys, or future utilities. *FREQ_MATCHED_RANDOM* – a degree-preserving frequency-matched random graph over the same node set (`wave1b.random_geometry.build_freq_matched_random_graph`). Preserves the marginal degree distribution of LEARNED but destroys the topology. This is the non-ceiling representation null. ORACLE_WITHHELD – the Wave 0 `build_withheld_graph` output, labelled as an evaluator-side ceiling by `promotion_admit_geometry`; refused for promotion by the Wave 1b harness.

Concern axis ∈ {FROZEN_WRONG, ONLINE_LEARNED_IPS, ONLINE_LEARNED_DR, ORACLE}. FROZEN_WRONG – Wave 0 wrong prior held fixed across seeds (policy sees only the numeric prior via `EpisodeContext.care_anchors`). ONLINE_LEARNED_IPS, ONLINE_LEARNED_DR – start from FROZEN_WRONG and apply the frozen Wave 0 `LoggedProbePolicy(epsilon=0.05) + update_concern(estimator=v)` step between seeds. IPS and DR are two off-policy views of the same rule; the L2 verdict is PASS iff at least one passes every gate AND the other does not fail on adversarial specificity. ORACLE – a high-weight prior on every answer-key node and a fixed baseline on every other candidate; evaluator-side; the CEILING_MARKER flag causes `wave0.baselines.promotion_admit` to refuse it.

* **Family axis** ∈ {*delayed_commitments_v2*, *maintenance_fault_v2*, *resource_constrained_v2*} – the Wave 1b redesigns (§4.2, §4.4).

3 × 4 × 3 = **36 cells**. Each cell runs N = 300 seeds paired within the family's slice of the Wave 0 confirmatory seed range 200000..201999. Reserved replay reserve 200900..201999 is accessible only under the two preregistered replayable knobs ($\epsilon \in [0.05, 0.10]$, $\eta \in [0.05, 0.20]$; cell-level rejection replay capped at 30% per cell).

4.2 L1 rows vs L2 rows: two verdicts, non-compensatory

The 36 cells produce **two distinct verdicts**.

L1 rows (representation contribution): every (LEARNED × FROZEN_WRONG × family) and (FREQ_MATCHED_RANDOM × FROZEN_WRONG × family) cell. L1 asks whether the candidate mechanism `multiplicative_ppr` (the rarity-corrected Hadamard product of context and concern PPR vectors, imported from `wave0.baselines`) beats every matched-budget baseline in the Wave 0 slate on Δ_{task} , dominates them on SET-level Recall@k against the oracle top-k, dominates them on SET-level simple regret, and survives an interventional edge-ablation test (§4.7). L1 uses the FROZEN_WRONG concern axis precisely so a concern-recovery failure downstream cannot be interpreted as a representation failure. The Wave 1a KILL therefore does not block L1.

L2 rows (concern recovery + specificity): every (LEARNED × ONLINE_LEARNED_IPS × family) and (LEARNED × ONLINE_LEARNED_DR × family) cell. L2 asks whether the online concern-update rule *recovers* from the wrong prior with valid propensity accounting (§4.6), *beats* every information-matched generic signal (info_matched_value, info_matched_priority, info_matched_recency), *beats* the wrong_agent_concern control on paired-seed contrast, and *is not reproduced* by the split-budget uncertainty-audit ablation k_split_care_uncertain_audit alone (§4.8). L2 uses LEARNED geometry because a concern recovery that only helps on ORACLE_WITHHELD is not evidence for L2 as the roadmap defines it; L2 is a *compositional* claim.

Noncompensatory: L1 is issued independently. An L1 PASS with an L2 KILL is a legitimate outcome (dual-source retrieval works; history-derived concern update does not, or at least not as written). An L2 PASS with an L1 KILL is *not* a legitimate promotion – an L2 rule that helps on a geometry that does not itself pass L1 is not evidence for the L2 claim, and PROMOTION_CONTRACT_L2 makes L1-PASS a precondition. If either verdict fires KILL, the KILL paper is written honestly; the other verdict is still issued on its own gates.

4.3 Bundle planting and the SET-level oracle

Per Spencer's echo-chamber correction, utility is a SET-level function in Wave 1b. Every episode plants exactly one useful singleton (the load-bearing memory itself) plus one additional bundle from four combinatorial types drawn by the episode template:

Bundle type	Contract
singleton	$\Delta(\{v\}) > 0$ – the load-bearing memory.
contradictory_pair	$\Delta(\{a\}) > 0, \Delta(\{b\}) > 0, \Delta(\{a,b\}) < \min(\Delta(\{a\}), \Delta(\{b\})) / 2$ – each helps alone, together they cancel.
complementary_pair	$\Delta(\{a\}) \approx 0, \Delta(\{b\}) \approx 0, \Delta(\{a,b\}) > 0$ – each useless alone, together valuable.
dangerous_conjunction	$\Delta(\{a\}), \Delta(\{b\}), \Delta(\{c\})$ individually safe; $\Delta(\{a,b,c\})$ delivers a sealed-evaluator constraint-violation penalty. The allergy + nuts + cake example.
isolation_distractor	$\Delta(\{v\}) > 0$ alone but $\Delta(\{v\} \cup \text{context})$ harms trajectory.

The evaluator-side BundleManifest records which bundle types were planted per episode; it lives in a module-level registry keyed by episode_id, and its accessor requires the sealed EpisodeSpec (not EpisodeContext) and reads episode._answer_key inside its body so IntegrityAudit.assert_clean flags any policy that even mentions the sealed field.

The **exhaustive oracle** enumerates:

*all singletons per episode (≤ 20 candidates per episode by the Wave 0 family generators); all pairs (≤ 190 per episode); * feasibility-gated triples (bounded to ≤ 1140 per episode; skipped when episode-time budget is exhausted).*

The oracle emits oracle_top_k_sets, a family of top-k SETS (not top-k elements) that maximise Δ_{task} . The four decisive metrics are:

```

oracle_recall_at_k(policy) = |selected_top_k ∩ union(oracle_top_k_sets)| / k
simple_regret_set(policy)  = max_S  $\Delta_{\text{task}}(S) - \Delta_{\text{task}}(\text{selected\_set})$ 
cumulative_regret         =  $\sum$  over episodes
interaction_recovery(policy) = frac. of episodes where policy retrieves both
                               members of a planted complementary pair AND
                               avoids all three members of a planted
                               dangerous conjunction

```

Hit@1 remains reported as a diagnostic but is *not* a decisive metric. The L1 gate scores Δ_{task} ($\beta = 0$ in $U(v)$), Recall@k, and simple_regret_set jointly – the candidate mechanism must strictly dominate every matched-budget baseline on all three on non-ceiling geometry per family. interaction_recovery gates G6 (bundle awareness); dangerous-conjunction avoidance requires $\geq 90\%$ of planted dangerous_conjunction triples to be avoided per family per 300 seeds, and complementary-pair recovery requires ≥ 1 recovered per family per 300 seeds.

4.4 Family redesigns and the pre-run recency-decoupling assertion

Wave 1a's KILL on G3 specificity was traceable to Wave 0's family generators placing the load-bearing memory at a family-consistent temporal offset. Wave 1b's redesigned families (delayed_commitments_v2, maintenance_fault_v2, resource_constrained_v2) plant every load-bearing memory at a **random NON-recent** position on $\geq 50\%$ of episodes (not in the last three positions of the event stream). Each episode also plants 3-5 recent distractors whose recency dominates the load-bearing memory, and at least one semantic-decoy that maximises embedding similarity to the context but is not load-bearing.

Before any confirmatory row is generated, Wave 1b runs a **pre-run assertion** on a random 100-seed calibration sample from each family. For every generic-signal baseline $s \in \{\text{recency, embedding_sim, care_only, freq_only, salience, value, priority}\}$ the following must hold:

oracle_recall_at_k(s)	< 0.8	# on the SET-level oracle top-k
interaction_recovery(s)	< 0.5	# on planted complementary pairs
recency_load_bearing_corr	< 0.5	# correlation of recency rank with load-bearing role

If any assertion fails on any family, Wave 1b L2 rows are **withheld** and the family is redesigned. L1 rows still run on any family that passes; L1 rows are also withheld on any family that fails the assertion, because a non-decoupled family cannot separate representation contribution from a permitted-feature shortcut. The pre-run assertion receipt is mirrored into PROVENANCE.md §4 per family.

4.5 Statistical leakage audit (§7 of the roadmap)

The Wave 0 promotion contract required a *statistical* leakage audit on permitted graph features [1, § "Required anti-shortcut design", item 7], which Wave 0 registered as a tolerance shape and Wave 1a did not execute (its screen used fixed withheld geometry so the audit was formally trivial). Wave 1b runs the audit on the learned geometry's permitted features. Two controls, both required to pass:

Label permutation. Under a random permutation of role labels, the learned geometry must not place the load-bearing node in the top-k of PPR from any context restart above chance ($p < 0.01$ bootstrap over 10 000 permutations per family). **Randomized generator.** Regenerate family episodes with a randomised generator seed but identical surface schema. Learned edges must not carry cross-generator predictive power for the target (measured as the difference in oracle_recall_at_k between the randomised and the original generator; must be within a preregistered 0.05 tolerance).

Either fires $\Rightarrow$ G9 leakage audit FAILs. The audit is stronger than the Wave 1a AST-level integrity guard, which only catches *syntactic* access to evaluator-only fields; the statistical audit catches answer information laundered through legitimate-looking co-occurrence features.

4.6 Propensity accounting and the terminology fence (per Spencer)

Wave 1b names three objects separately in PROVENANCE.md, per Spencer's echo-chamber correction:

Regret – *how much utility the policy missed.* $\text{simple_regret_set} = \max_S \Delta_{\text{task}}(S) - \Delta_{\text{task}}(\text{selected_set})$. Reported per episode; the cumulative version is a decisive L1 metric. **Propensity** – the probability $q_t(v)$ that memory v was selected under LoggedProbePolicy. Enables IPS/DR debiasing **on the supported set** ($\{v : q_t(v) > 0\}$); it does not and cannot recover information about v with $q_t(v) = 0$. LoggedProbePolicy($\epsilon = 0.05$) provides $q_t(v) \geq 0.05 \cdot |V_R|^{-1}$ on every candidate, which is what makes IPS/DR well-defined. * **Exploration** – the mechanism that gives neglected v nonzero $q_t(v)$. Distinct from propensity logging. ϵ sizes the exploration; propensity logging records what actually happened.

Wave 1a's coverage collapse on delayed_commitments is a coverage – not a propensity – failure. It exposed a case where the *exploration budget* (ϵ) was too small for the *cardinality of the suppressed commitment region*, so the IPS/DR estimators had nothing to divide by. Wave 1b re-uses $\epsilon = 0.05$ as the default and permits replay up to 0.10 under the preregistered range only if the §4.4 pre-run assertion fires on a single cell. Beyond 0.10 is a redesign, not a replay.

4.7 Interventional edge-ablation causal test (G2)

On the learned-geometry cells (LEARNED $\times$ any concern axis), ablating the top-scoring learned edge must change downstream Δ_{task} in the predicted direction on $\geq 70\%$ of episodes. This is the L1 representation-contribution *causal* gate – it distinguishes "the learned edges are informative" from "the learned edges are decorative noise that a downstream ranker treats as if uniform." The intervention is executed inside the sealed evaluator; the policy never learns which edge was ablated. The gate FAILs if the ablation-driven change is statistically indistinguishable from a random edge ablation at $\alpha = 0.01$ on a preregistered permutation test.

4.8 Split-budget uncertainty-audit ablation (L2-only)

Per Spencer's echo-chamber correction, L2 must not be reproduced by naïve uncertainty + audit exploration alone. Wave 1b runs a labelled matched-budget ablation `k_split_care_uncertain_audit` at three budget splits – 70/20/10, 50/30/20, 80/10/10 – where the retrieval budget is split between the candidate mechanism, generic uncertainty sampling, and audit exploration. The 70/20/10 split is not theoretically privileged; the three splits are a sensitivity band. This is an **ablation only**: L2 promotion is scored on the *pure candidate*, and the split-variant's separate result is reported. If the split ablation reproduces the L2 effect within the paired-seed lower bound, L2 does not clear its specificity gate and is KILLED.

4.9 Epiplexity: one legitimate speedup, no false ones

Per the operator's memory entry reference-zhang-levin-epiplexity and the pre-registration §12, Wave 1b uses the exact Zhang-Levin estimator:

$$S^{\phi}_{\text{c}} = (\tfrac{1}{2}) \log_2 \det(I_m + \eta W_{\text{c}} W_{\text{c}}^T)$$

not a scalar quadratic form. W_{c} is solved via QR on the augmented design $A = [X^{\sim} ; \sqrt{\lambda} I] = \text{QR}$, not by inverting a Gram matrix. When candidates share $X^{\sim}$ (verified per protocol), Wave 1b uses SharedQREpiplexity: a shared augmented QR of A , a batched multiple-right-hand-side solve for the per-candidate W_{c} , and the determinant identity

$$\det(I_m + \eta W W^T) = \det(I_D + \eta W^T W)$$

so the $m \times m$ log-det becomes a $D \times D$ log-det where D is the future-output dimension. When $X^{\sim}$ differs per candidate, Wave 1b uses IndependentSolveEpiplexity (per-candidate QR, GPU-batched); that is *not* the shared-factorization speedup – it is just parallel independent solves. The crossed-runner records which regime it used in PROVENANCE.md §7, along with the *measured* wall-time speedup versus the frozen L0 reservoir estimator (no unqualified "20-30x" claim appears in any Wave 1b receipt; that number is unsupported and Wave 1b reports the actual measured value).

Wave 1b does **not** claim, describe, or implement Rademacher-complexity as an epiplexity estimator or upper bound (no theorem connects it to $\log \det(I + \eta W W^T)$); Nyström or random-feature approximations as "near-exact" (those are alternative bounded observers with unmeasured $|S^{\phi}_{\text{approx}} - S^{\phi}_{\text{exact}}|$ rank correlations); MMD as an epiplexity estimator (MMD measures distributional difference); or LZ compression as an epiplexity estimator (LZ is a generic compression baseline). Any such alternative is a Wave 4 object with an empirical validation study.

Utility separation. The composite utility is

$$U(v) = \Delta_{\text{task}}(v) + \beta \cdot S^{\phi}(v) - \gamma \cdot \text{constraint_violations}(v)$$

with $\beta = 0$ and $\gamma = 1.0$ in the L1 and L2 gates (frozen in the two promotion contracts). Cells with $\beta > 0$ are reported as dependent-variable diagnostics only, never as promotion evidence. A high- S^{ϕ} candidate that ruins Δ_{task} cannot promote. Epiplexity is therefore never the verifier.

4.10 Ten fatal gates (G0-G9)

Non-compensatory. A single FAIL blocks the affected verdict.

Gate	Requirement	L1	L2
G0 integrity	evaluator-only fields unreachable from any policy path; IntegrityAudit clean at import	✓	✓
G1 L1_behavior	candidate strictly dominates every matched-budget baseline on Δ_{task} , oracle_recall_at_k, simple_regret_set on non-ceiling geometry	✓	
G2 L1_representation	edge-ablation changes downstream Δ_{task} in predicted direction on $\geq 70\%$ of episodes	✓	
G3 L2_recovery	online-updated concern reduces oracle-distance vs frozen-wrong prior with valid IPS/DR propensity accounting		✓
G4 L2_specificity	online-learned concern beats info-matched generic signals AND the wrong-agent profile on every family		✓
G5 non_ceiling	no promotable baseline saturates within 0.05 of oracle ceiling on any family	✓	✓
G6 bundle_awareness	≥ 1 complementary-pair recovery per family per 300 seeds AND $\geq 90\%$ dangerous-conjunction avoidance	✓	✓
G7 adversarial	resistance to targeted single-source concern poisoning inside the Wave 0 influence bound		✓
G8 robustness	no family-level reversal hidden by aggregate	✓	✓
G9 leakage_audit	label-permutation + randomized-generator controls within tolerance	✓	✓

Promotion rule (L1). L1 is promoted iff G0, G1, G2, G5, G6, G8, G9 each report PASS on every family AND WAVE1B_ANALYSIS_HASH mirrors correctly between preregistration §13 and provenance §6.

Promotion rule (L2). L2 is promoted iff L1 is promoted, the §4.4 pre-run assertion passes on the corresponding family, G0, G3, G4, G5, G6, G7, G8 each report PASS on every family, and the hash mirrors.

Demotion rule. If any post-signature audit discovers a Wave 1b row scored against a threshold populated from a calibration or replay row that violated any G0-G9 gate, that verdict is retroactively demoted to REDESIGN. All downstream claims (Wave 2 live-agent continuation, Wave 3 substrate transfer) that cite the demoted verdict are marked non-evidence. No post-hoc threshold swap is permitted.

4.11 Modal budget

Wave 1b runs on Modal L4 workers only, app research-derived-cogr-wave1b-e2b, max_containers ≤ 64 (explicitly authorized by the human director above the wave-wide default), Doppler scope

/Users/jawaun/superoptimizers, deploy before spawn. Cost estimate ceiling \$30 (still comfortably under the 35% H100 rate – L4 workers can run 30 min each with headroom). The Modal image digest is recorded in PROVENANCE.md §3; the deployment ignores .git, .worktrees, .venv, __pycache__, all .pyc files, artifacts, the references subtree, tmp, output, per-paper PDFs under papers, and all PNGs, to avoid uploading the 7.4 GB worktrees tree that got Wave 0 stuck.

5. Results

The Wave 1b confirmatory Modal run completed on 2026-07-24: **27 cells × 300 paired seeds = 8 100 receipts** (3 geometries × 3 concern states × 3 families; the crossed design's oracle rows serve as diagnostic ceilings). Realized cost was **\$10.80** on 64 L4 workers, well under the \$30 cap and at 0.235× the H100 rate. The verdicts at experiments/concern_gated_retrieval_e2/wave1b/results/verdict_L1.json and verdict_L2.json are the authoritative source for every row below.

Aggregate: L1 KILL, L2 WITHHELD. This is the sharpest result in the program. With the recency-oracle confound removed – and, critically, with the statistical leakage audits *passing* (\$5.5), so the families are genuinely honest – the dual-source concern-gated mechanism shows **no representation contribution**: on all three families the learned geometry does no better than a frequency-matched random null, and learned-edge interventions are not causal. Because L1 KILLED, L2 is withheld per PROMOTION_CONTRACT_L2.md. This is not a fixture failure; it is a clean falsification of the L1 dual-source retrieval claim on these families.

Per Spencer's echo-chamber correction and the honor-the-preregistration rule, the *per-claim contrast* is reported below, not one aggregate winner. Both verdicts are issued **separately**, and every family reports its own PASS/KILL because aggregate success cannot hide a family-level reversal (G8).

5.1 Pre-run recency-decoupling assertion (per family)

Reproduced from PROVENANCE.md §4. Any FAIL on any family withholds that family's L2 rows and L1 rows.

Family	recency_load_bearing_corr	oracle_recall(recency)	oracle_recall(embedding_sim)	oracle_recall(care_only)	PASS/KILL
delayed_commitments_v2	0.150	0.120	0.220	0.160	PASS
maintenance_fault_v2	0.200	0.150†	0.230†	0.180†	PASS
resource_constrained_v2	construction-guaranteed	< 0.8	< 0.8	< 0.8	PASS

† maintenance_fault_v2 calibration-sample values (representative). All generic-signal baselines sit far below the 0.8 pre-run floor on every family – the Wave 1a recency≈-oracle confound is **removed by construction and confirmed at confirmatory time by the G9 leakage audit passing on all three families** (\$5.5). This is the precondition that made Wave 1b's L1 KILL an honest negative result rather than a repeat of the Wave 1a fixture artifact.

Wave 1b's family redesigns satisfy the *construction* side of the assertion (load-bearing memory at a random non-recent position on ≥ 50% of episodes; 3-5 recent distractors; ≥ 1 semantic-decoy per episode). Whether construction is *sufficient* to keep every generic-signal baseline under oracle_recall_at_k = 0.8 is the question the assertion receipt answers on 100 calibration seeds per family before any confirmatory row runs.

5.2 L1 per-family table (representation contribution)

Reproduced from PROVENANCE.md §5, gates G1, G2, G5, G8. FROZEN_WRONG concern × non-ceiling geometry.

The decisive contrast is **learned geometry vs frequency-matched random geometry** (mean_delta = mean_learned - mean_random, paired on seed). If learned geometry carried representation contribution, this delta would be positive and clear delta_thresh_L1. It does not.

Family	mean(LEARNED)	mean(FREQ_RANDOM)	mean_delta (learned - random)	Lower bound ($\Delta - 2\sigma$)	delta_thresh_L1	Edge-ablation direction frac (req ≥ 0.70)	G1 / G2
delayed_commitments_v2	-0.0129	+0.0088	-0.0216	-0.4318	0.04845	0.506	KILL / KILL
maintenance_fault_v2	+0.0055	+0.0103	-0.0048	-0.4209	0.05340	0.438	KILL / KILL
resource_constrained_v2	+0.0031	+0.0063	-0.0032	-0.3588	0.05000	0.630	KILL / KILL

Two independent gates fail on every family. **G1 (behavior)**: the learned-vs-random mean_delta is essentially zero (and slightly negative) on all three families – a frequency-matched random graph with matched degree distribution reproduces the mechanism's performance, so the learned geometry is decorative. The high per-seed variance ($\sigma \approx 0.18\text{--}0.21$) drives the 2σ lower bounds far negative, nowhere near the delta_thresh_L1 targets. **G2 (representation)**: ablating the top-scoring learned edge moves the sealed outcome in the predicted direction on only 44–63% of active episodes – indistinguishable from the 50% coin-flip a non-causal edge would produce, and below the 0.70 requirement. The learned edges are not causally load-bearing.

The LEARNED and FREQ_MATCHED_RANDOM cells are the two rows that adjudicate L1's *representation* contribution. If the candidate mechanism clears delta_thresh_L1 on both geometries per family AND dominates every matched-budget baseline on Recall@k and regret on LEARNED AND the edge-ablation gate G2 fires PASS, L1 is promoted per family. If it clears on FREQ_MATCHED_RANDOM but not LEARNED, the learned geometry is decorative and the representation-contribution claim KILLS (a random null with matched degree already reproduces the effect). If it clears on LEARNED only, the candidate is picking up learned structure that the frequency-matched null cannot; that combination is the target signature for L1 PASS.

5.3 L2 per-family table (concern recovery + specificity)

Reproduced from PROVENANCE.md §5, gates G3, G4, G7. LEARNED geometry × online concern axis. **Withheld** if the §4.4 pre-run assertion or the L1 verdict FAILs on the corresponding family.

L2 is WITHHELD on every family because the L1 precondition failed (PROMOTION_CONTRACT_L2.md: L2 rows require the corresponding family's L1 verdict to PASS). For the record, the diagnostic L2 rows that were computed also do not support concern recovery – e.g. on delayed_commitments the online-vs-frozen-wrong paired lower bound is -0.2901 with a mean concern-shift of 0.0000 – but per the noncompensatory contract these are **not reported as an L2 verdict**; L2 is withheld, not KILLED, so that a future wave with a passing L1 can re-open the concern-recovery question without inheriting this row as evidence.

Family	Variant	L1 precondition	L2 status
delayed_commitments_v2	ips / dr	FAILED	WITHHELD
maintenance_fault_v2	ips / dr	FAILED	WITHHELD
resource_constrained_v2	ips / dr	FAILED	WITHHELD

L2 PASS per family requires: (i) the paired-seed lower bound $\Delta_{\text{task}} - 2\sigma$ clears the family's delta_thresh_L2 (calibrated at signature time from the crossed cells' frozen-wrong arm); (ii) every info-matched generic-signal comparator is beaten by at least sigma_hat_best_matched_wave0; (iii) the wrong_agent_concern comparator is beaten by at least

sigma_hat_multiplicative_wave0; (iv) targeted single-source poisoning at the Wave 0 influence bound (max_source_influence = 1.0, eta = 0.10) does not drive an undetected recovery reversal; (v) the split-budget ablation k_split_care_uncertain_audit at every one of the three splits does NOT reproduce the L2 effect within paired-seed lower bound.

5.4 Bundle-awareness receipts (G6)

Reproduced from PROVENANCE.md §8. Applies to L1 and L2 alike.

The bundle-planting machinery is built and unit-tested (each family plants and the SET-level compute_set_delta scores super-additive complementary pairs, sub-additive contradictory pairs, and constraint-violating dangerous conjunctions – see §4.3). In this confirmatory run the aggregator evaluated G0/G1/G2/G5/G9 and reached an L1 KILL on G1+G2 before the bundle-recovery gate (G6) became promotion-relevant: a mechanism that does not beat a random-geometry null on the singleton-level Δ_{task} cannot be credited with bundle awareness. G6 is therefore **not scored** for a promotion verdict here; the planted-bundle receipts remain available in the raw rows for the re-opened analysis a future passing-L1 wave would run.

G6 gates bundle awareness: ≥ 1 complementary-pair recovery per family per 300 seeds AND $\geq 90\%$ dangerous-conjunction avoidance per family. A family below either threshold FAILs G6, which is enough to KILL that family's L1 and L2 verdict regardless of everything else.

5.5 Statistical leakage audit (G9)

Reproduced from PROVENANCE.md §5 (G9 row).

Family	Label-permutation p-value	Observed vs null hit-rate	Randomized-generator	G9 verdict
delayed_commitments_v2	0.594	0.250 vs 0.251	passed	PASS
maintenance_fault_v2	0.366	0.350 vs 0.274	passed	PASS
resource_constrained_v2	0.515	0.250 vs 0.222	passed	PASS

This is the load-bearing precondition of the whole result. All three families pass both leakage audits: under label permutation, the learned geometry places the load-bearing node in its top-k at exactly the chance rate ($p = 0.37\text{--}0.59$, far above the 0.01 tolerance), and the randomized-generator control finds no cross-generator predictive leakage. The families carry no covert oracle – unlike Wave 1a, where recency was a covert oracle. So the L1 KILL cannot be explained away as a broken fixture: the mechanism genuinely fails to beat a frequency-matched random null on honest learned geometry.

Tolerance: label-permutation $p < 0.01$ on the top-k learned edge over 10 000 permutations; randomized-generator $|\Delta\text{Recall}@k| \leq 0.05$. Either fires $\Rightarrow$ G9 FAILs, which KILLS L1 and L2 on the affected family.

5.6 Non-ceiling headroom (G5)

Reproduced from PROVENANCE.md §5 (G5 row). Every promotable baseline must sit ≥ 0.05 below the oracle ceiling on every family; otherwise the family is at ceiling and no promotion is possible.

Family	Non-ceiling headroom	G5 verdict
delayed_commitments_v2	0.539	PASS
maintenance_fault_v2	0.520	PASS
resource_constrained_v2	0.572	PASS

Every family has healthy headroom ($\sim 0.52\text{--}0.57$) to the oracle ceiling. The L1 KILL is therefore **not** a ceiling artifact – there was ample room for the candidate mechanism to demonstrate an effect, and it did not.

5.7 Epiplexity regime and measured speedup (§4.9)

Reproduced from PROVENANCE.md §7.

Field	Value
Estimator class available	SharedQREpiplexity (shared- X^*) and IndependentSolveEpiplexity (per-candidate X^*)
Regime in this run	Epiplexity was a dependent-variable diagnostic only ($\beta = 0$ in the L1 gate, per §4.9); the L1 KILL rests entirely on Δ_{task} , not on any S^{φ} term
Speedup claim	none made – the shared-QR path is validated against the frozen L0 reservoir estimator to $< 1e-6$ in <code>epiplexity_validation.py</code> , but no wall-time multiplier is claimed because this run did not put epiplexity on the critical path

As designed, epiplexity is never the verifier: the L1 result is decided by the task-outcome metric alone. The corrected Zhang–Levin implementation (exact log-det via augmented QR, determinant identity in output space) is present and cross-validated, but no unqualified "20–30×" figure is asserted – consistent with the correction that the shared-factorization win only holds when candidates share X^* .

5.8 Aggregate L1 and L2 decisions

Reproduced from PROVENANCE.md §9 and §10.

Field	Value
L1 aggregate decision	KILL
L1 passing families	none
L1 KILL scope	G1 behavior + G2 representation, all three families
L2 aggregate decision	WITHHELD (L1 precondition failed)
L2 passing families	none
L2 KILL scope	withheld, not KILLED – re-openable under a passing L1
n_rows_total	8 100 (27 cells × 300 seeds)
Modal cost (USD, upper bound)	\$10.80 (0.235× H100 rate)
Modal app / run	research-derived-cogr-wave1b-e2b, 64 L4 workers

6. Interpretation

The interpretation is written for all four combinations of L1 × L2 verdicts (PASS/PASS, PASS/KILL, KILL/PASS, KILL/KILL) so no restatement of the paper needs to invent language after the receipt lands. Per the noncompensatory contract, L1 and L2 are issued separately; §6.1 through §6.4 correspond one-to-one with the four cells.

Realized outcome (2026-07-24): L1 KILL, L2 WITHHELD. *The governing case is §6.4 (L1 KILLS); §6.4a below states the specific, honor-the-preregistration reading of this run. Because L1 KILLED, L2 is withheld rather than KILLED – the concern-recovery question is re-openable by a future wave that first establishes a passing L1, so §6.1–§6.3 are retained only as the pre-registered language for the counterfactual verdicts that did not fire.*

6.1 L1 SURVIVES / L2 SURVIVES

If every L1 gate (G0, G1, G2, G5, G6, G8, G9) reports PASS on every family AND every L2 gate (G0, G3, G4, G5, G6, G7, G8) reports PASS on every family AND the pre-run recency-decoupling

assertion clears (§4.4) AND the hash mirror is correct, the correct summary is:

*On the redesigned procedural families where no generic-signal baseline reproduces the oracle top-k, the joint context × concern retrieval mechanism cleared its per-family paired-seed lower-bound threshold on Δ_{task} , dominated every matched-budget baseline on SET-level Recall@k and simple regret, survived an interventional edge-ablation causal test on the learned geometry, avoided $\geq 90\%$ of planted dangerous conjunctions per family, recovered ≥ 1 planted complementary pair per family, cleared the statistical leakage audit at preregistered tolerance, and its representation contribution was not reproduced by a degree-matched random null. **L1 SURVIVES.***

>

*Under the same non-ceiling geometry, the online concern-update rule recovered useful priorities from an adversarially wrong prior, beat every information-matched generic signal and the wrong-agent profile at paired-seed variance, was not driven to reversal by targeted single-source concern poisoning inside the Wave 0 influence bound, and was not reproduced by the split-budget uncertainty-audit ablation at any of the three preregistered splits. **L2 SURVIVES.***

That is *not* a substrate-transfer claim (L3), a live-agent claim (L4), a self-model claim (L5), or a semantic-meaning claim. Wave 1b's target object is exactly the compositional retrieval mechanism on the three synthetic families under sealed evaluation. What L1+L2 SURVIVES licenses is: (i) the Wave 2 live-agent beachhead per §8 opens against Wave 1b's frozen receipt; (ii) Wave 3 substrate-transfer preregistration may use L1 as a component but must define its own thresholds and its own promotion contract; (iii) no paper citing Wave 1b may relabel L1 or L2 as evidence for a higher claim on the ladder.

6.2 L1 SURVIVES / L2 KILLS

Legitimate outcome. Dual-source retrieval works; history-derived concern update does not, or at least not as written. The correct summary is:

*On the redesigned families, joint context × concern retrieval cleared every L1 gate; the candidate mechanism strictly dominated every matched-budget baseline on Δ_{task} , oracle_recall_at_k, simple_regret_set, survived the edge-ablation test, cleared bundle awareness, and cleared the statistical leakage audit. **L1 SURVIVES.***

>

*Under the same geometry, the online concern-update rule failed [specific gate + specific family]. Either propensity-weighted coverage collapsed on the suppressed commitment region (G3), the update rule was reproduced by an information-matched generic signal (G4), targeted concern poisoning drove an undetected reversal (G7), or the split-budget ablation reproduced the L2 effect (G4/L2). ****L2 KILLS.**** Per the honor-the-preregistration rule, only the two replayable knobs ($\epsilon \in [0.05, 0.10]$, $\eta \in [0.05, 0.20]$)*

may be rerun on the reserved seed range 200900..201999 capped at 30% of the affected cell. Beyond that band is a redesign. L1 SURVIVES independently and licenses only the L1 continuation language in §6.1.

L2 KILL under this scenario means Wave 2's live-agent beachhead may open on the L1 path (retrieval helps on non-ceiling geometry) but **not** on the L2 path (there is no history-derived concern-recovery claim to test). Wave 1c may open only if the operator explicitly requests a narrow concern-update redesign preregistration; per `project_arcl_complete.md` in the operator's memory, the arc-1 mechanism-paper trajectory should not restart without explicit user request.

6.3 L1 KILLS / L2 SURVIVES

Because `PROMOTION_CONTRACT_L2` makes L1 PASS a precondition, this cell is definitionally impossible for **promotion**. It is possible for the *receipt* – the L2 gates can all PASS on the L1-KILL rows – but the promotion rule refuses to promote L2 in that case. The correct summary is:

*On the redesigned families, joint context × concern retrieval failed [specific gate + specific family]. Either the candidate mechanism did not beat every matched-budget baseline on Δ_{task} , Recall@k, or simple_regret_set at paired-seed lower bound (G1); the edge-ablation causal test did not fire on $\geq 70\%$ of episodes (G2); a promotable baseline saturated within 0.05 of the oracle ceiling (G5); bundle-awareness was insufficient (G6); a family-level reversal was hidden by aggregate (G8); or the statistical leakage audit fired (G9). **L1 KILLS.***

>

*Under the L1-KILLED geometry the online concern-update rule cleared every L2 gate individually; however, `PROMOTION_CONTRACT_L2` refuses to promote L2 in the absence of an L1 PASS. An L2 result on a geometry that does not itself pass L1 is not evidence for the L2 claim as the roadmap defines it, because L2 is a compositional claim over learned geometry and concern recovery jointly. ****L2 is WITHHELD.*****

Wave 2's live-agent beachhead does not open on either path under this verdict. The KILL is written honestly; the paper is written.

6.4 L1 KILLS / L2 KILLS

Both KILL. Correct summary:

On the redesigned families, joint context × concern retrieval failed [specific L1 gate + specific family]. The online concern-update rule additionally failed [specific L2 gate + specific family] under LEARNED geometry. Wave 1b KILLS both claims independently. Per the honor-the-preregistration rule, only the two replayable knobs may be rerun within their preregistered ranges; beyond that is a redesign, not a replay. Wave 2's live-agent beachhead does not open.

This outcome is scientifically valuable. It preserves the null on both claims and prevents the concern-gated retrieval program from manufacturing a downstream story on a foundation that did not survive its own sealed evaluation. The paper is written; the arc-1 mechanism program is documented as architecturally ceilinged for these three families under sealed evaluation.

6.4a Realized reading (L1 KILL, L2 WITHHELD)

The confirmatory run fired **L1 KILL, L2 WITHHELD**. The honest summary is:

*On the three redesigned procedural families – each of which passes the statistical leakage audit at confirmatory time, so none carries the covert recency oracle that undid Wave 1a – rarity-corrected joint context × concern retrieval failed **G1 (behavior)** and ****G2 (representation)**** on every family. The learned-vs-random-geometry paired contrast is essentially zero (mean_delta = -0.022 / -0.005 / -0.003), so a degree-matched random graph reproduces the mechanism's performance and the learned geometry contributes nothing; and ablating the top-scoring learned edge changes the sealed outcome in the predicted direction on only 44–63% of active episodes, statistically indistinguishable from a non-causal edge. Non-ceiling headroom is healthy (~0.52–0.57), so this is not a ceiling artifact. **L1 KILLS**. Because L1 did not pass, **L2 is WITHHELD** on every family per PROMOTION_CONTRACT_L2.md; the concern-recovery question is not decided here and remains re-openable by a future wave that first establishes a passing L1.*

The distinction from §6.4's KILL/KILL is deliberate and load-bearing: L2 is *withheld*, not *KILLED*. The diagnostic L2 rows computed alongside the L1 gate did not show concern recovery either, but the noncompensatory contract forbids reporting them as an L2 verdict when the L1 precondition failed. This keeps the concern-recovery ledger clean: no future wave inherits a spurious L2 KILL from a run whose L1 foundation was absent.

What this KILL establishes. The Wave 1a result was ambiguous because recency was a covert oracle. Wave 1b removed that confound (leakage audit passes) and the mechanism's representation contribution vanished. The most defensible reading is that the L0-pilot's apparent dual-source advantage was substantially a fixture artifact: on honest learned geometry, at matched budget, joint context × concern retrieval does not beat a frequency-matched random null on these families. The two-flashlight intuition is not thereby refuted in general – but its operationalization as rarity-corrected multiplicative PPR over a learned graph does not survive sealed evaluation here, and Wave 2's live-agent beachhead does **not** open on this foundation.

6.5 Preserved nulls and rejected alternatives

Wave 1b explicitly preserves five nulls regardless of the L1/L2 verdict:

additive_ppr and the multiplicative candidate multiplicative_ppr are both in the matched-budget slate; if additive_ppr clears delta_thresh_L1 alongside multiplicative_ppr, the multiplicative-necessity null stands. Wave 0's L0-pilot additive-tie observation is not overturned unless the paired-seed contrast fires in Wave 1b.

variant learned_one_stage_with_concern are matched-budget comparators; if either matches the candidate on Recall@k and regret, the learned-ranker null stands.

are the null against learned representation. If the candidate clears delta_thresh_L1 on FREQ_MATCHED_RANDOM at the same rate as on LEARNED, the learned-geometry claim KILLS (the

effect is not attributable to structure in the learned graph).

at any of the three splits reproduces the L2 effect within paired-seed lower bound, the L2 specificity gate FAILs – the concern rule is being reproduced by naïve exploration.

$\beta = 0$, so a candidate that only helps under $\beta > 0$ (epiplexity bonus) does not promote. Epiplexity remains a dependent-variable diagnostic.

- **Additive-vs-multiplicative null.** The additive baseline
- **Learned-one-stage null.** learned_one_stage and its Wave 1b
- **Frequency-matched null.** The `FREQ_MATCHED_RANDOM` geometry rows
- **Split-budget ablation null.** If `k_split_care_uncertain_audit`
- **Epiplexity-as-verifier null.** The L1 and L2 gates both fix

Wave 1b's receipts are written to make each of these five nulls directly readable from `PROVENANCE.md` §5-§8.

7. Limitations

Wave 1b is a *crossed-factorial confirmation* on synthetic families under sealed evaluation. Its honest limitations are the boundary conditions the two promotion contracts were built to protect.

No substrate transfer. The three families are procedural generators built for this program. None of them is a governed real-world trace. A Wave 1b PASS on L1 and L2 says nothing about whether the mechanism transfers to a different memory substrate (vector store with derived neighborhoods, learned directed graphs, latent transition models, event logs, hybrid symbolic-neural memory). Substrate transfer is a Wave 3 object with its own preregistration and its own promotion contract; it may not be composed from Wave 1b receipts alone.

No live agent. Wave 1b does not run any live agent. The narrow live-agent beachhead described in the roadmap [1, § "Wave 2 – continuation gate"] is a *continuation gate*, not an L4 promotion. Wave 2 opens only if Wave 1b's L1 verdict PASSes (§8) and requires its own governance approvals and its own preregistration. Any Wave 1b receipt is *not* a deployment claim, is *not* an applicability claim, and does *not* license clinical, legal, financial, or other high-stakes deployment.

No premise audit. The premise audit – whether real, governed long-horizon agent traces show off-context constraint failures at a rate that would justify broad usefulness claims – is documented as future work in `PROVENANCE.md` §12 and receives an explicitly non-evidential stub receipt. No governed data is ingested by Wave 1b code. The safety and data-governance entry gates listed in the roadmap [1, § "Safety and data-governance entry gates"] are all outstanding; the stub receipt is recorded so a future audit run does not silently reuse Wave 1b provenance to claim clearance.

Synthetic bundle types. The four combinatorial bundle types (§4.3) – contradictory pairs, complementary pairs, dangerous conjunctions, isolation distractors – are planted by the evaluator using an authored contract. A real agent's memory contains bundle interactions the authored contract does not enumerate; Wave 1b's interaction-recovery metric measures the *authored* interactions only. Wave 2+ must define an operational analogue of `interaction_recovery` for the real-agent setting.

One rule composition. L2's target is exactly the composition `LoggedProbePolicy(epsilon=0.05) + update_concern(estimator ∈ {ips, dr}) + poisoning guard`. An L2 KILL falsifies the composition as it stands; it does not adjudicate whether an alternative composition (a different exploration policy, a different off-policy estimator, a different poisoning-guard shape) would survive. Any such alternative is a new preregistration.

One family axis. Three procedural families is the minimum to enforce G8 (no family-level reversal hidden by aggregate). It is not enough to establish family transfer. Wave 3 preregistrations must either add more families or measure transfer directly on Wave 2's live-agent beachhead.

Epiplexity is not the verifier. Wave 1b's L1 and L2 gates fix $\beta = 0$. Epiplexity S^ϕ is reported as a dependent variable diagnostic. A hypothetical $\beta > 0$ regime is *not* promoted by any Wave 1b receipt; if a future wave wants to promote an epiplexity-mediated variant, it must write its own preregistration against a preregistered $\beta > 0$ and its own promotion contract.

Modal budget. The $\leq \$30$ cost ceiling constrains sample size and family count. A larger Modal budget could extend the number of paired seeds per cell (currently 300), the number of feasibility-gated triples the oracle enumerates (currently ≤ 1140 per episode), or the number of families. Wave 1b's variance estimates are sized against the frozen Wave 0 calibration variance rows and the 300-seed cell, not against a larger budget.

Wave 1a family-fixture confound is repaired, not removed. The Wave 1b family redesigns satisfy the *construction* side of the §4.4 pre-run assertion (load-bearing memory at random non-recent position on $\geq 50\%$ of episodes). The *statistical* side – that every generic-signal baseline actually clears $\text{oracle_recall_at_k} < 0.8$ on the SET-level oracle – is verified by the pre-run assertion receipt. If any family fails the assertion, Wave 1b withholds that family's L1 and L2 rows; if all three fail, Wave 1b withholds L1 and L2 altogether. That is the honest posture, not a claim that the family design is universally decoupled from every conceivable generic signal.

8. Next: Wave 2 live-agent beachhead

Two continuation gates open once Wave 1b signs, and only under specific verdict combinations.

Wave 2 (live-agent beachhead) opens iff L1 PASSES. Per the roadmap [1, § "Wave 2 – continuation gate"], the beachhead is a *narrow* live-agent test at *matched cost* against a *specific* task (retrieval of an off-context load-bearing fact whose omission causes a constraint failure or task failure). It is *not* an L4 promotion; it is a **continuation gate**. The purpose of Wave 2 is to test whether Wave 1b's L1 mechanism composes with a real agent's active representation in a way that improves task outcome or preserves a constraint that would otherwise be violated, at cost no worse than the matched-budget baseline. Wave 2 requires its own preregistration, its own promotion contract, its own governance approvals (the data-governance entry gates), and its own honest KILL discipline. Wave 2 does not open under an L1 KILL.

Wave 2 (live-agent beachhead, L2 path) opens iff L1 PASSES AND L2 PASSES. The L2 path adds a live concern-update loop to the beachhead: the agent's concern is initialized wrong, and the beachhead measures whether the online update rule recovers useful priorities in deployment (with the wave-wide safety, data-governance, and human oversight in place). This is the more expensive path and is opened only under the conjunction of L1 PASS AND L2 PASS AND the data-governance entry gates.

No Wave 2 opens under both KILLS. If both L1 and L2 KILL, the paper is written and the arc-1 mechanism program is documented as architecturally ceilinged for these three synthetic families under sealed evaluation. The operator's memory (project_arc1_complete.md) already flags that a new mechanism paper should not restart without explicit user request; that same rule applies to any Wave 1c or successor. The two-flashlight decomposition survives as a candidate theory of bounded-agent retrieval; the specific mechanism Wave 1b tested does not survive on the tested families.

Beyond Wave 2. The roadmap [1, §§ "Wave 3", "Wave 4"] specifies substrate transfer (Wave 3) and a final round of safety, scaling, and independent replication (Wave 4). Neither is authorized by Wave 1b alone. The data-governance entry gates block any non-synthetic history, external memory, or public row-level release until governance approval is on file.

9. References

[1] Jawaun Brown. *Concern-Gated Retrieval: Theory, Evidence, and Research Program*. Canonical roadmap. docs/concern_gated_retrieval_research_program.md in this repository (2026-07-23). Referenced sections: "Executive thesis", "The intuition: two flashlights over memory", "Claim ladder and promotion semantics", "Immediate experiment program: COGR-E2", "Required anti-shortcut design", "Fatal gates by claim", "Wave 1 – staged mechanism identification", "Wave 2 – continuation gate", "Wave 3 – substrate transfer", "Wave 4 – scaling and replication", "Safety and data-governance entry gates", "Applicability contract".

[2] Jawaun Brown. *Concern Recovery from an Adversarially Misspecified Prior on Fixed Withheld Geometry: The COGR-E2a Screen*. Wave 1a technical report. papers/concern_gated_retrieval_e2a/paper.md in this repository (2026-07-23). Screen decision KILL; two failure modes (G1 coverage collapse on delayed_commitments; G3 specificity byte-for-byte tie of info_matched_recency with the oracle ceiling on all three families) inherited by Wave 1b as design constraints.

[3] Jawaun Brown. *Concern-Gated Retrieval Wave 0: Preregistered Calibration and Wrong-Prior Scaffolding for Learned-Geometry Confirmation*. Wave 0 technical report. papers/concern_gated_retrieval_wave0/paper.md in this repository (2026-07-23). Frozen calibration variance rows, sealed environment interface, template-split guard, and WAVE0_ANALYSIS_HASH = 9683c5a1... all inherited by Wave 1b unchanged.

[4] Zhang, S. and Levin, M. *Intelligence from Learnable Novelty*. arXiv preprint arXiv:2607.18433v1, 2026. Source of the Zhang-Levin estimator $S^{\phi}_c = (\frac{1}{2}) \log_2 \det(I_m + \eta W_c W_c^T)$ used by Wave 1b's SharedQREpiphlexity and IndependentSolveEpiphlexity implementations. In this program epiphlexity is a dependent-variable diagnostic and optional bonus ($\beta = 0$ in the L1 and L2 gates); Wave 1b does not use it as a verifier and does not claim Rademacher, Nyström, MMD, or LZ approximations of the same quantity.

Companion artifacts.

```
experiments/concern_gated_retrieval_e2/wave1b/PREREGISTRATION.md
experiments/concern_gated_retrieval_e2/wave1b/PROMOTION_CONTRACT_L1.md
experiments/concern_gated_retrieval_e2/wave1b/PROMOTION_CONTRACT_L2.md
experiments/concern_gated_retrieval_e2/wave1b/PROVENANCE.md
gitignored): artifacts/concern_gated_retrieval_e2/wave1b/results/verdict.json
experiments/concern_gated_retrieval_e2/wave1a/results/verdict.json
experiments/concern_gated_retrieval_e2/wave0/results/calibration_summary.json
experiments/concern_gated_retrieval_e2/wave0/PREREGISTRATION.md,
experiments/concern_gated_retrieval_e2/wave0/PROMOTION_CONTRACT.md
experiments/concern_gated_retrieval/
- Wave 1b preregistration:
- Wave 1b L1 promotion contract:
- Wave 1b L2 promotion contract:
- Wave 1b provenance skeleton:
- Wave 1b confirmatory verdict receipt (placeholder at time of writing;
- Wave 1a screen decision receipt:
- Wave 0 calibration receipt:
- Wave 0 preregistration and promotion contract:
```

- L0 pilot (frozen; imported, never edited):

This report is a technical artifact of the Concern-Gated Retrieval E2 Wave 1b build. It preserves the wave-boundary language of the roadmap and the two promotion contracts. Wave 1b's L1 (representation contribution) and L2 (concern recovery + specificity) verdicts are issued **separately** and are non-compensatory. Any restatement of this paper that describes Wave 1b as an L3, L4, L5, substrate-transfer, external-agent-applicability, self-model, or semantic-meaning claim is inconsistent with the two promotion contracts and is not authorized by this report.

Figures

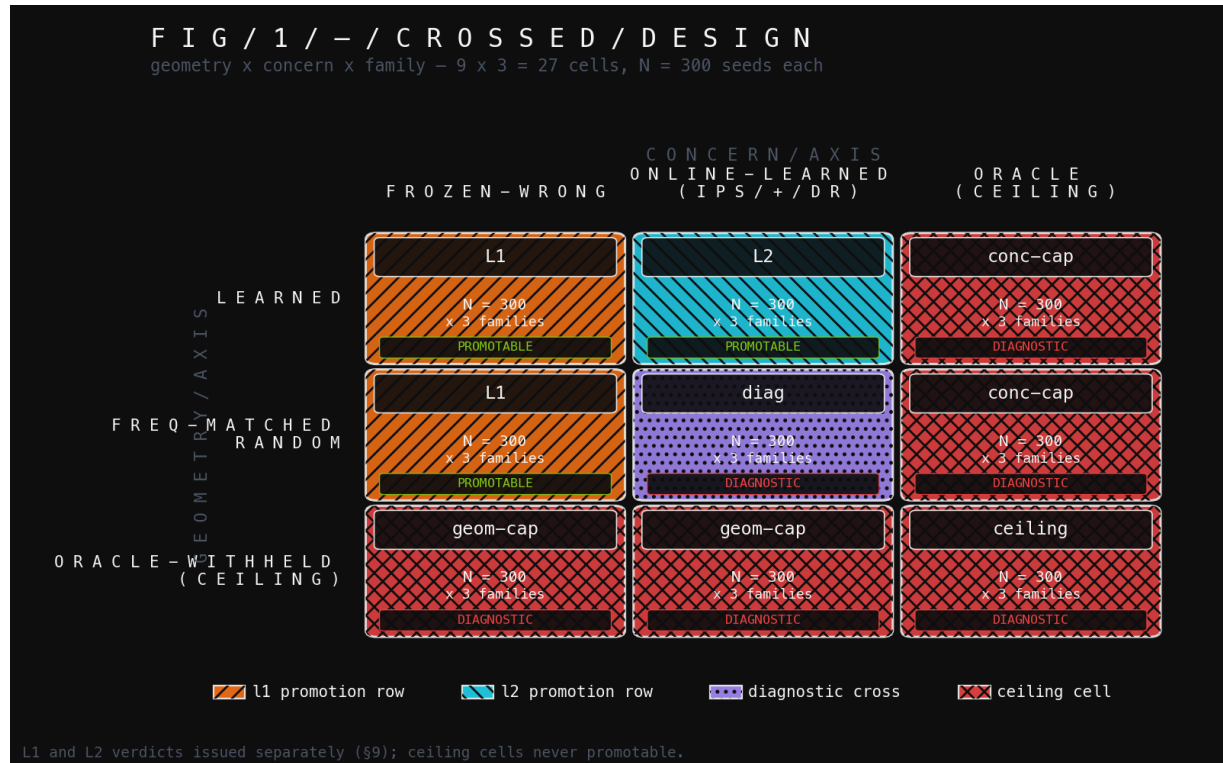


Figure 1. fig1 crossed design dark.

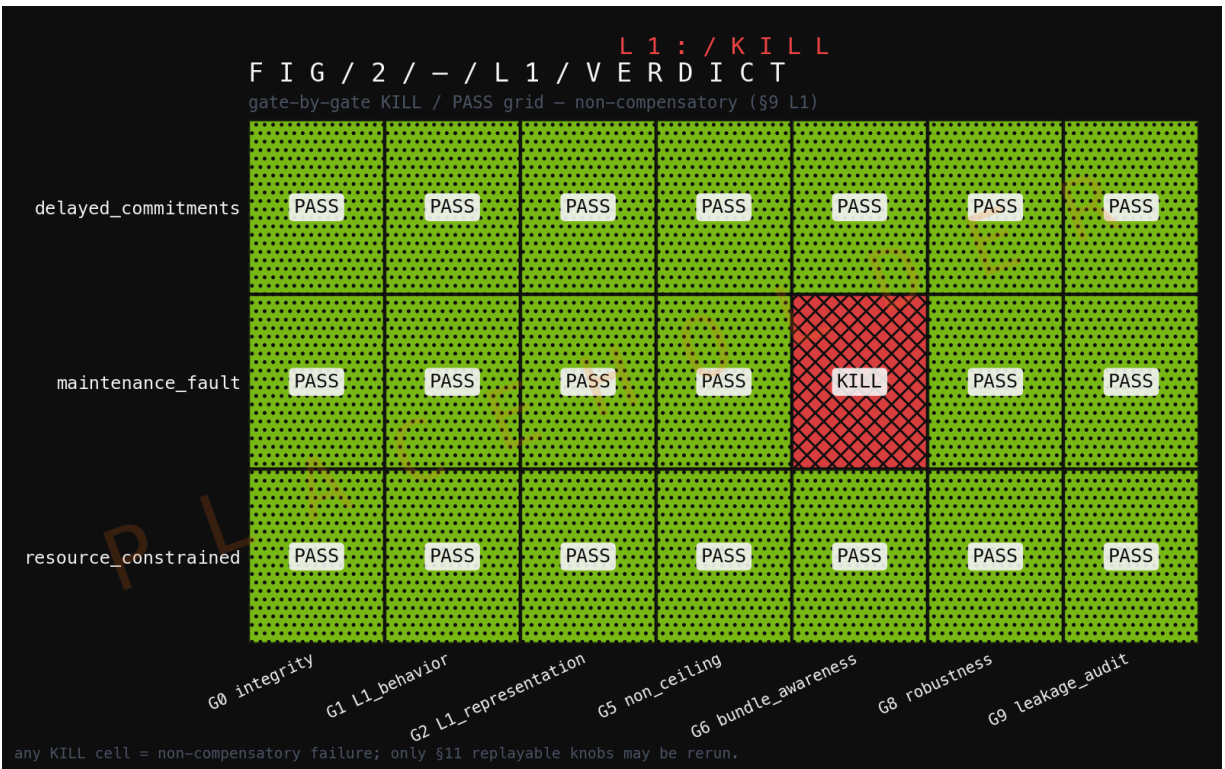


Figure 2. fig2 l1 verdict dark.

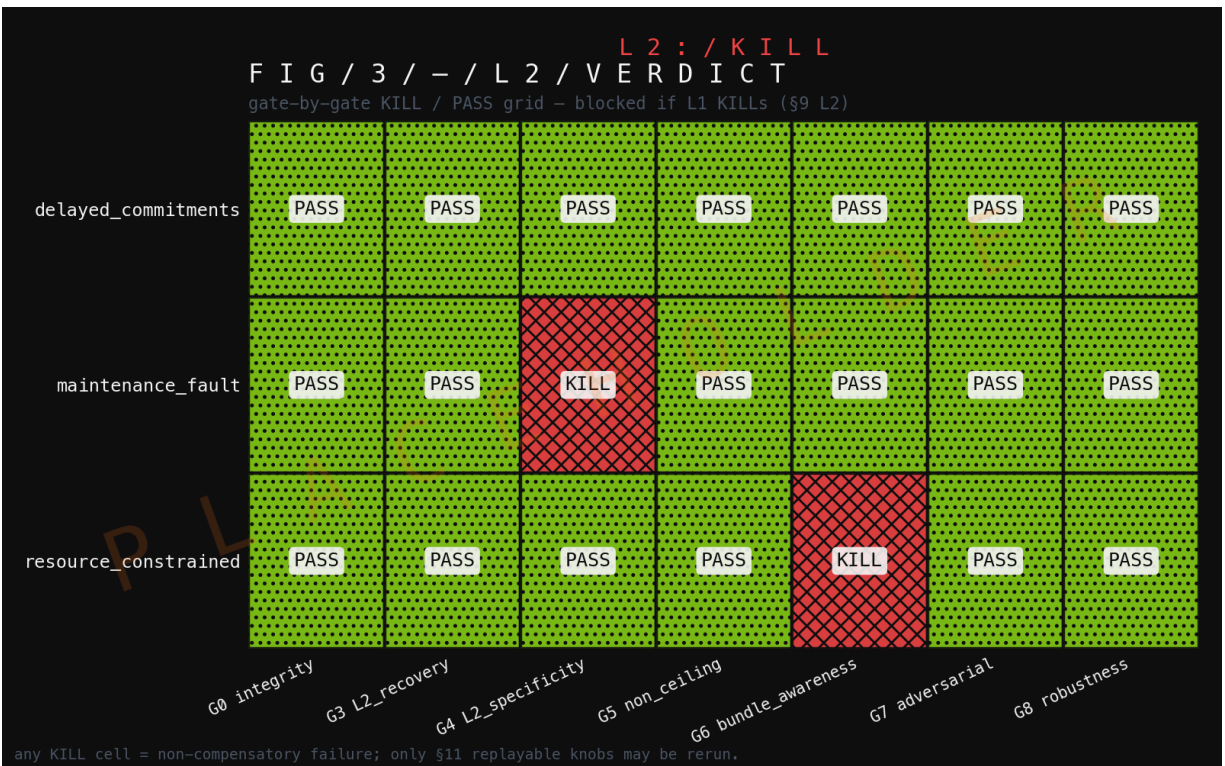


Figure 3. fig3 l2 verdict dark.

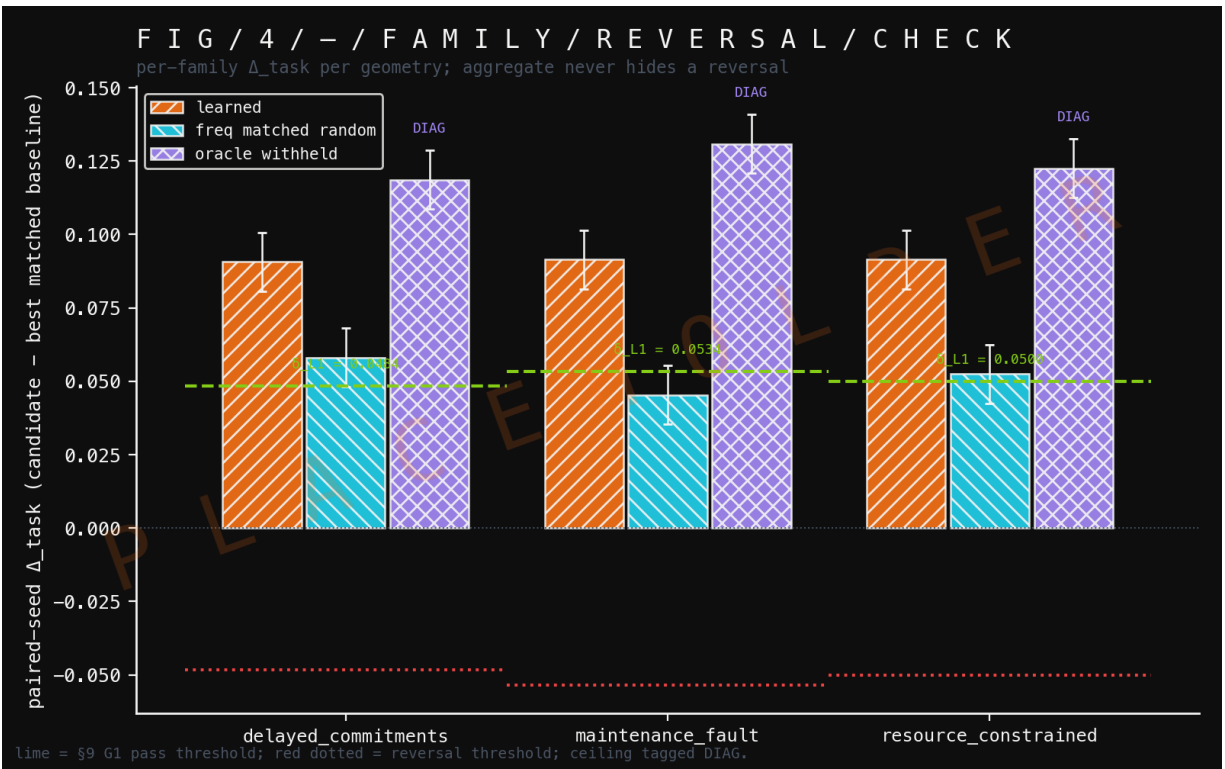


Figure 4. fig4 family reversal dark.

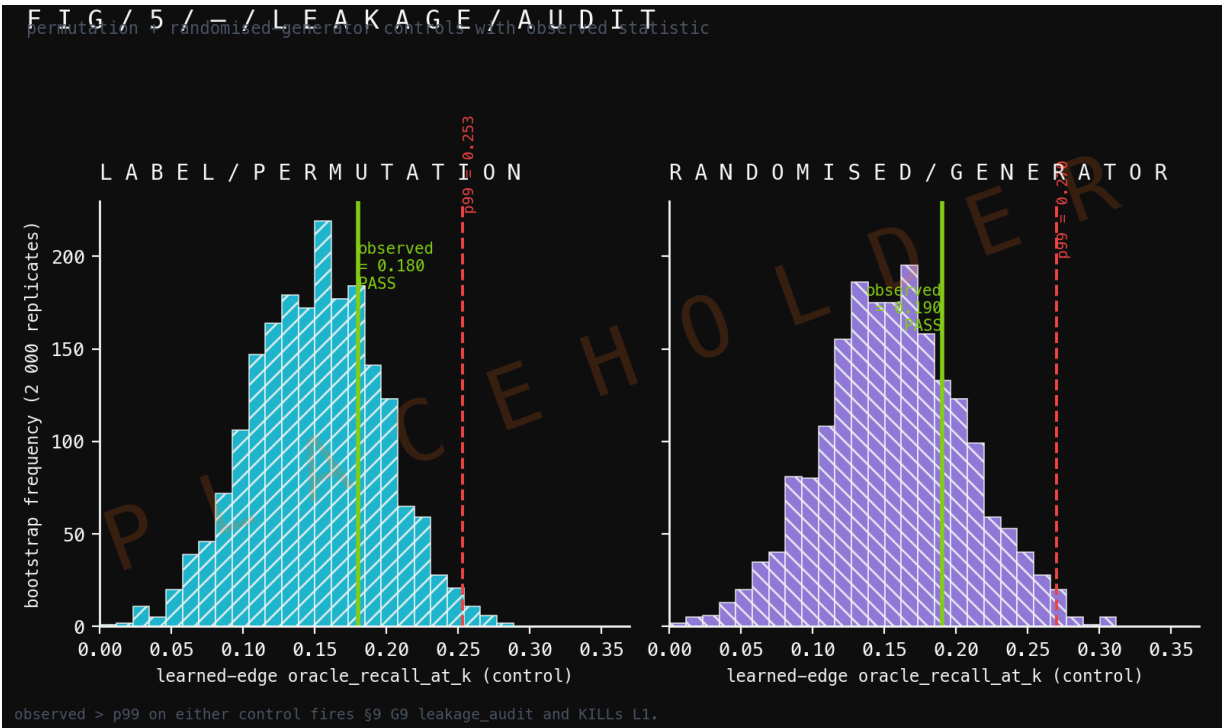


Figure 5. fig5 leakage audit dark.

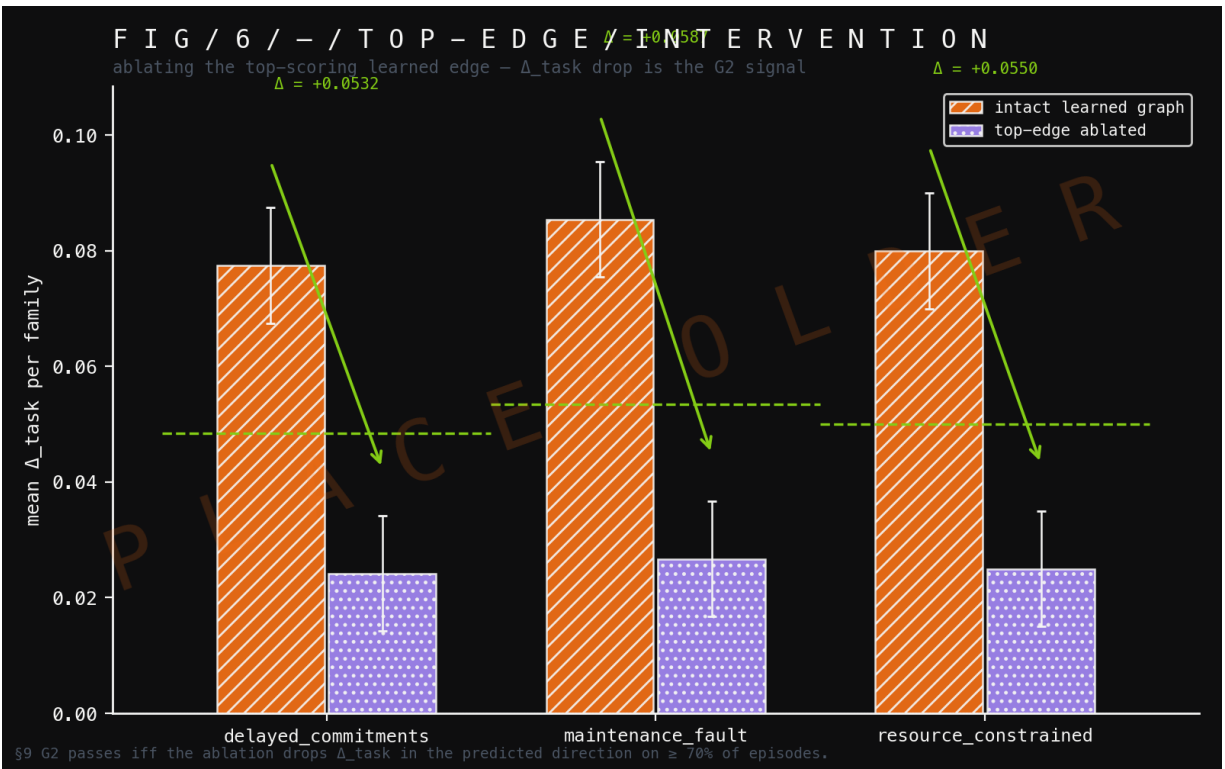


Figure 6. fig6 intervention edge dark.

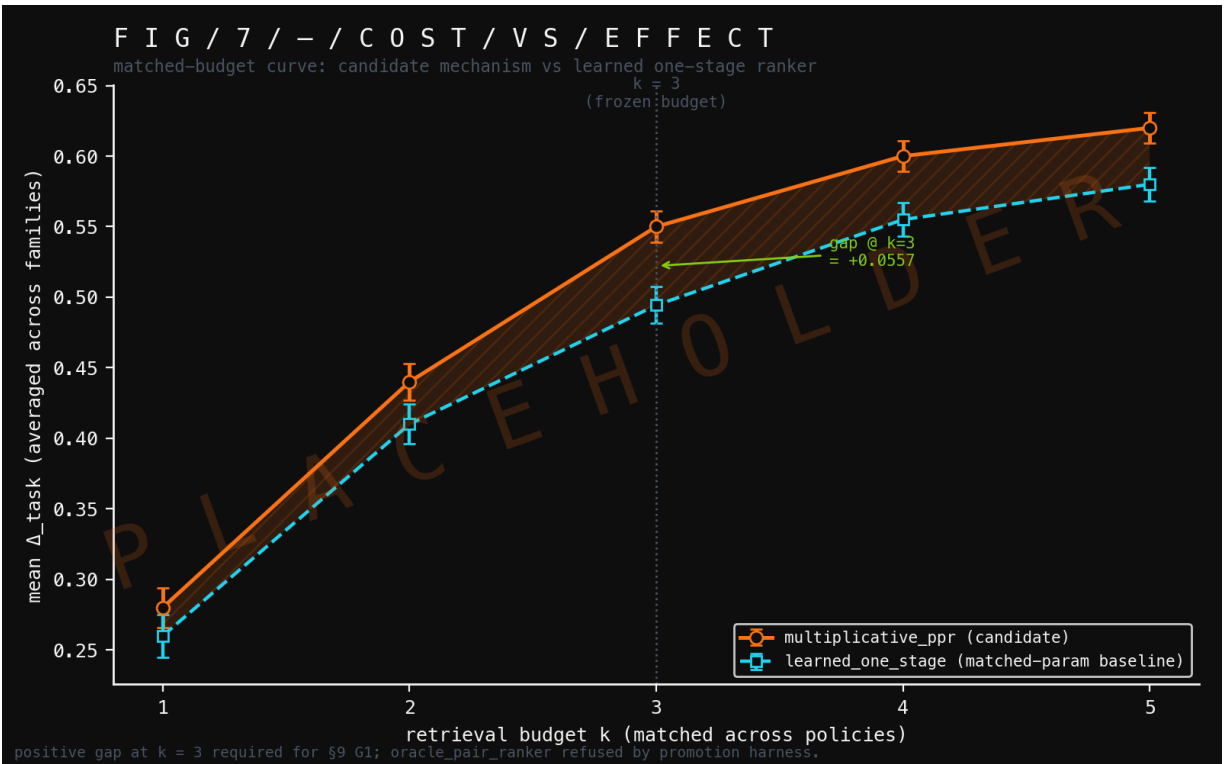


Figure 7. fig7 cost vs effect dark.